

Feb. 24,
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PA2 - Network flow

A: incoming x_1 , outgoing x_2 , ext. flow +100 inc.
 $x_1 = x_2 - 100$

B: incoming x_2 , 50, outgoing x_3
 $x_2 + 50 = x_3$

C: incoming x_3 , outgoing x_4 , ext. flow -120 exit
 $x_3 = x_4 + 120$

D: incoming x_4 , outgoing x_5 , ext. flow -150 exit
 $x_4 = x_5 + 150$

E: incoming x_5 , outgoing x_6 , 80
 $x_5 = x_6 + 80$

F: incoming x_6 , 100, outgoing x_1 .
 $x_6 + 100 = x_1$

- ① $x_1 - x_2 = -100$, ② $x_2 - x_3 = -50$,
③ $x_3 - x_4 = 120$, ④ $x_4 - x_5 = 150$,
⑤ $x_5 - x_6 = 80$, ⑥ $x_6 - x_1 = -100$

$$\begin{bmatrix} 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 \\ -1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{matrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{matrix} = \begin{bmatrix} -100 \\ -50 \\ 120 \\ 150 \\ 80 \\ -100 \end{bmatrix}$$